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Application & Scope Management in ServiceNow

What is an Application? (The Deployment Container)

An **Application** is the final, distributable package of your configurations. For an Admin, this means moving hundreds of changes (forms, scripts, ACLs) as **one single file** instead of managing dozens of Update Sets.

- An Application solves the **deployment headache**. You use it to install, upgrade, or uninstall a large feature set cleanly, guaranteeing **atomicity** (all files move or none move).
- Before deployment, an Admin confirms all necessary files (artifacts) have been correctly associated with the correct application scope.
 - **Action:** Verify the sys_scope field value on new configuration records (e.g., a Business Rule or Table).
- If an artifact is accidentally created in the global scope, it will be missed during application deployment, causing the app to partially fail in Production.

Use Case:

A customer needs a new **Contract Management** system. Instead of using 15 Update Sets to move the changes, the Admin publishes the **Contract Management App** to the **App Repository**.

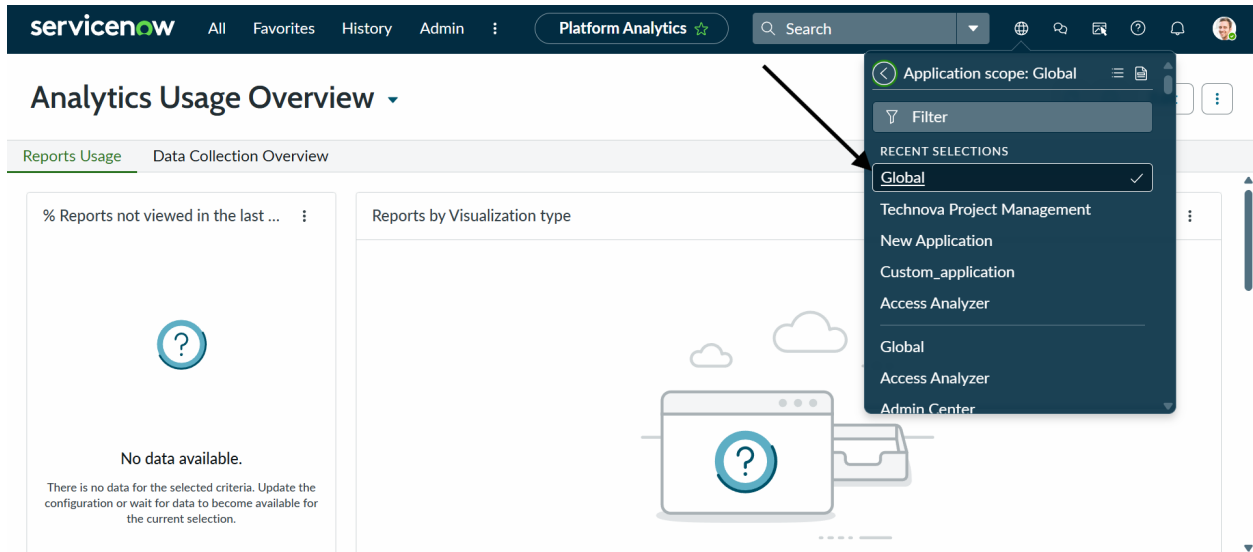
- **Analogy:** The Application is a **Zip File**. You don't manage every file inside the folder; you just move the single, complete zip file.

[**Note:** Update Sets are like “change packages” that capture configuration changes made in the Global scope, so they can be moved from one instance to another (e.g., Dev → Test → Prod)]

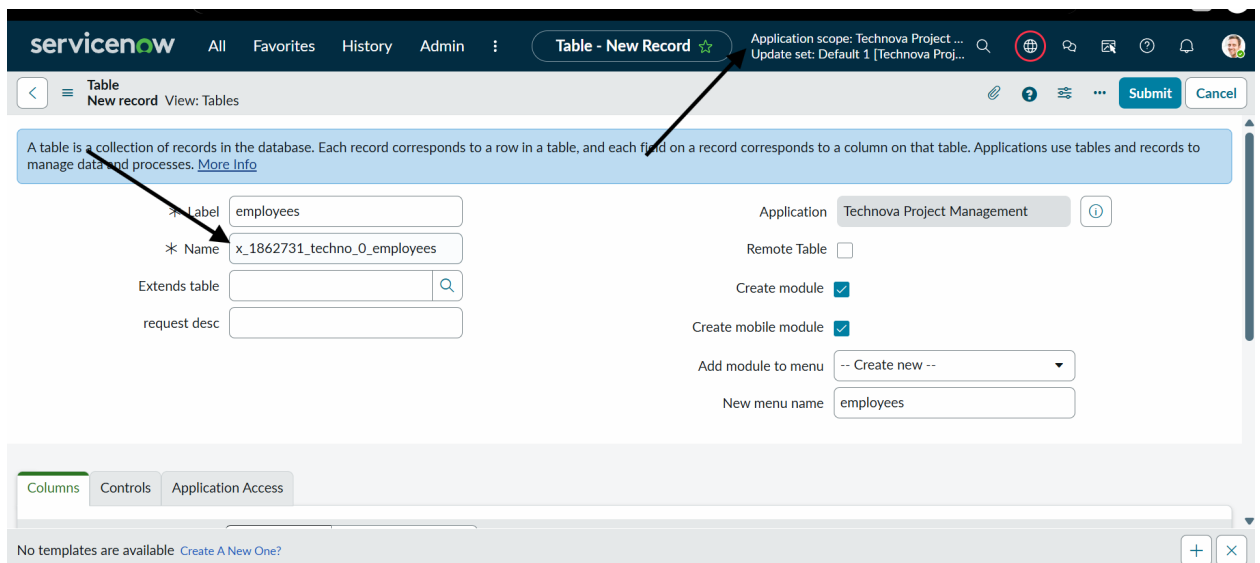
What is Application Scope?

Scope is the active workspace boundary. For an Admin, the Scope picker dictates **where new records go** and **what you are allowed to edit**.

- The first rule of creating anything new is: **Check the Application Picker**. If you are building a custom feature, the picker **must** be set to the corresponding Scoped App, not Global.
- **Action:** Switching the **Application Picker** (header dropdown) from Global to your custom scoped application.



- Creating a table while in the wrong scope means the table will not be packaged with the correct application, breaking your deployment strategy.
- The moment you switch scope, the system automatically prepends the unique scope prefix (e.g., **x_acme_**) to all new tables and fields you create.



- **Table Name Check:** Always confirm your custom table name starts with the correct scope prefix (e.g., x_acme_safety_audit).
- **Switching Scope:** While in Studio, the scope is automatically set. When working in the main UI (e.g., creating a new Form Layout), always use the **Application Picker**.

Use case:

Admin Use Case: An Admin is tasked with modifying a field on the **Incident** table (Global) and then creating a new **Vacation Request** table (Scoped). They must switch scopes twice.

- **Analogy:** The Scope is your **Keycard**. You use the *Global* keycard to access the IT Service Management floor, and the *HR* keycard to access the HR Application floor. You can only work where the keycard is valid.

Global vs. Scoped Applications

The choice between Global and Scoped is the most critical administrative decision that determines the fate of the application lifecycle.

Decision Point	Global Scope (ITSM Mods)	Scoped Application (New Process)
When to Use	Modifying OOB tables (incident, change), or " Glue Code " (integrating multiple systems).	Building NEW , distinct business processes (Legal, Facilities, Custom HR).
Deployment Method	Update Sets (Manual, prone to error).	App Repository (Recommended, automated update/install).
Security Risk	High: All scripts can modify any OOB record.	Low: Data is protected by scope boundaries (RCA).
Table Prefix	Custom tables/fields use u_ .	Custom tables/fields use x_[code]_ .

Admin Action	Risk Assessment: Assess impact of changes on core processes.	Application Creation: Must use Studio/GAC to establish the scope first.

{ **Note: "Glue code"** in the context of ServiceNow usually refers to code that connects it with other systems, such as the data integration tool AWS Glue, or custom scripts that enable interoperability between different parts of a ServiceNow instance or other applications }

Deployment using scoped applications:

- **Global Consequence:** Requires meticulous **Update Set** management. Admins must manually ensure all files (e.g., the XML file for an ACL, the XML for a new Role) are included. If one file is missed, the feature breaks in Production.
- **Scoped Consequence:** Deployment is cleaner. The Admin simply publishes a new **Application Version** to the App Repository. The target instance (Prod) installs the version, ensuring all necessary files are included atomically.

How to Create a Scoped Application (3 Ways)

The creation process is initiated through three main methods, depending on the complexity required.

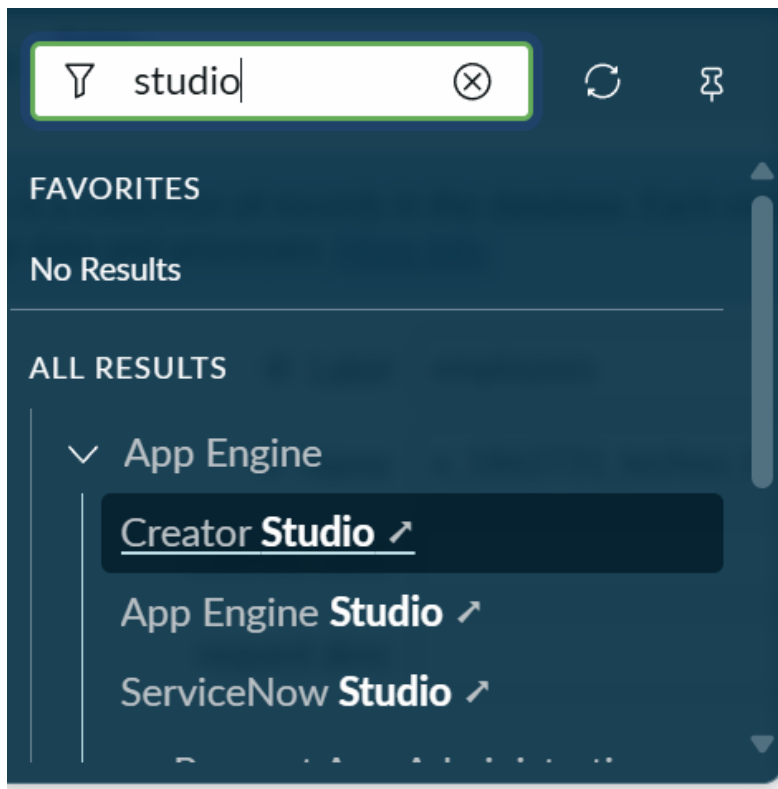
Method	User Focus	Navigation Path	Practical Admin Use
1. App Engine Studio (AES)	Citizen/Low-Code Developers.	App Engine → App Engine Studio	Starting point for all new apps. Provides guided experience for creating tables, roles, and security, and integrates with App Engine Management Center for pipeline deployment.

2. Guided App Creator (GAC)	Admins/Standard Developers.	System Applications → My Company Applications → Create new	A traditional wizard that defines the Application Shell (Name, Scope ID, Initial Table) and then immediately hands off to Studio.
3. ServiceNow Studio	Coders/Advanced Developers.	System Applications → Studio → Create Application	Used for advanced customization like creating apps from an existing template or importing from Source Control (Git) , bypassing the GAC wizard.

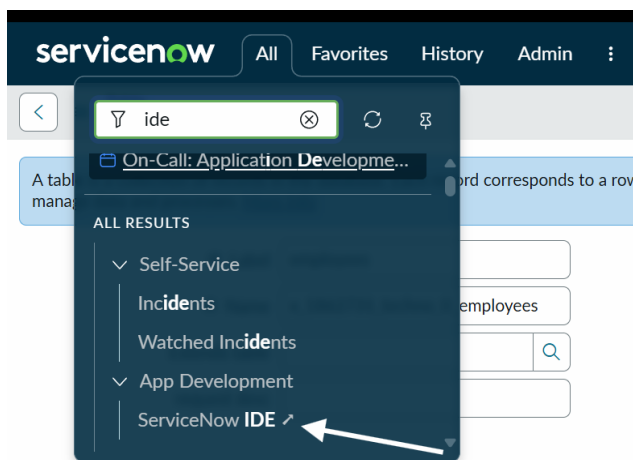
Comparison: Studio vs. Main UI

- **Use Studio For:** Creating new tables, designing Form Layouts, configuring security (ACLs), and writing scripts.
- **Use Main UI For:** Testing the app (creating actual records), importing data via Loaders, or running reports (as Studio doesn't have a record view).

{ **Preferred Practice:** Always create Scoped Applications using **ServiceNow Studio**, as it is the modern and recommended development environment. Studio provides better structure, improved security, file organization, and future-ready capabilities, making it ideal for ongoing and scalable application development. { **Note:** Try to create scoped applications using the ServiceNow studio, its latest way of creating applications and recommended for future development also }



{ Note: There is also other option for Application Development “ServiceNow IDE” which is an implementation of Visual Studio Code for the Web on the ServiceNow AI Platform }

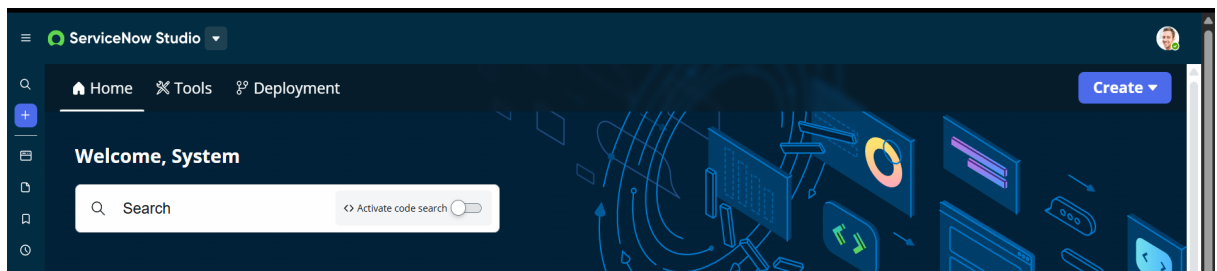


Step by Step process for creating Scoped Application using ServiceNow Studio:

The Studio method is used when importing an application from a **Git repository** or when the Admin wants full control, bypassing the GAC wizard.

Step-by-Step: Creating a New Application in Studio

1. **Launch Studio:** Type **studio** in the Filter Navigator or click **System Applications** → **Studio**.
2. **Initiate Creation:** Click the **Create Application** button.
3. **Choose Source:** Select **"Start from scratch"** (to define the app from basic fields) or **"Import from Source Control"** (to pull existing code from Git, used in CI/CD pipelines).
4. **Define Application Details (Start from scratch):** This is the crucial step where the application identity is established.

A screenshot of the 'CREATE APP' form in ServiceNow Studio. The form is titled 'Let's get started on your new app.' and includes instructions to add a name, description, and thumbnail. It features input fields for 'Name' and 'Description', a 'Scope' section with 'Scoped' and 'Global' radio buttons, and a 'Browse or drag to upload' area for a thumbnail image. A sidebar on the right promotes 'AI-powered Now Assist'. The bottom status bar shows 'Scope: None'.

Field Name	Description & Input	Admin Rationale (Why it Matters)

Name	The user-friendly title (e.g., " Legal Compliance Tracker ").	This becomes the Application Menu Label in the navigator.
Scope	(Auto-populated: x_[code]_[app_name]). Do not change.	This prefix guarantees uniqueness and applies the correct namespace to all future artifacts (tables, fields, scripts).
Description	A brief summary of the app's business purpose.	Helps future administrators and governance teams understand the application's function.
Version	Default is 1.0.0. Follow your organization's versioning standard (e.g., Semantic Versioning).	Essential for managing updates. The system tracks files against this version number during publishing.
Table	Option to create the primary table immediately (e.g., Compliance Case).	Saves time by linking the first core data structure directly to the new application scope.

Why Scoped Applications are More Secure

Scoped security provides layers of protection an Admin must configure and monitor.

Security Feature	Admin Configuration Point	Consequence of Failure
Cross-Scope Access (RCA)	Application Access tab on the Table record (sys_db_object).	An external script attempts to modify data but is blocked by the system's security manager.
Data Isolation	Setting Can Read/Write to false for "All Application Scopes."	Prevents read access to confidential data (e.g., HR data) from all other apps.
Delegated Development	Application Administration → Delegated Development	Granting non-admin users specific permissions (e.g., can only create UI Policies) within the app, reducing Admin workload and risk.
IP Protection	Setting Script Includes to Protected when publishing to the App Repo.	The customer cannot steal or modify the core logic of the application.

If a script in Global attempts to access a table in a Scoped App and fails, the Admin must:

1. Check the **System Logs** for the **Restricted Caller Access** failure message.
2. If access is desired, navigate to **System Applications** → **Application Restricted Caller Access** and create a new record to explicitly grant the Global scope the ability to Read/Write the specific target artifact.

Use case:

You need the Global Incident App to look up the **Team Lead** from your Scoped "Project Management" App. The Admin must go to the Project Management App's settings and grant a **Cross-Scope Read Privilege** to the Global scope.

Naming Conventions for Scopes

For Admins, the naming prefix is the unique ID. It determines deployment, uniqueness, and file location.

Prefix	Scope	Admin Action/Implication
x_[code]_	Scoped	The standard for all new custom applications. Mandatory for App Repo publishing.
u_	Global	Used only when creating custom tables or fields <i>outside</i> of a scope (e.g., modifying Incident). Avoid for new systems.
sn_	ServiceNow	Indicates the artifact belongs to a ServiceNow Store application. (Hands-off for Admin).

Implementation:

- **Company Code:** The [code] part is set by ServiceNow when the instance is provisioned and is fixed. The Admin has no control over it.
- **Naming Rule:** Keep the [AppName] section short and descriptive. The entire scope name must be under **40 characters**.
- If a table name shows up as u_x_acme_myapp_mytable, the Admin made a mistake: they were in the **Global** scope and tried to create a table with a scope prefix. The system treats the entire string as a global custom table name.

Use Case: When preparing a custom application for submission to a corporate internal repository, the Admin confirms the naming is x_acme_legal_docs. This prefix guarantees that when another internal team installs it, their configurations won't clash.

